

SIMATS ENGINEERING

ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 10%

QUESTIONS: 50

Total Marks:50

Annexure A MCQ Test

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Q.No	Question	Option A	Option B	Option C	Option D	Ans
1	Preprocessing improves feature detection by	A) Increasing size	B) Reducing noise	C) Changing format	D) Increasing brightness	
2	Without preprocessing, detection becomes	A) Better	B) Same	C) Inconsistent	D) Faster	
3	Grayscale conversion helps in	A) Increasing color	B) Reducing complexity	C) Increasing resolution	D) Adding noise	
4	Binary representation is useful for	A) Texture detection	B) Foreground separation	C) Color enhancement	D) Contrast increase	

5	Image representation affects	A) Pixel size	B) Feature visibility	C) Memory only	D) Edge removal	
6	Edge detection identifies	A) Noise	B) Intensity change	C) Smooth regions	D) Colors	
7	Canny is preferred due to	A) Speed	B) Low error	C) Memory	D) Size	
8	Sobel is sensitive to	A) Color	B) Noise	C) Size	D) Resolution	
9	Weak edges occur due to	A) High contrast	B) Low gradients	C) High resolution	D) Size	
10	Edge detection fails when	A) Smooth image	B) Strong gradients	C) Noise dominates	D) High resolution	
11	Corner detection finds	A) Flat regions	B) Edges	C) Intensity variation	D) Noise	
12	Harris is sensitive to	A) Rotation	B) Translation	C) Intensity	D) Noise	
13	Corners are useful because	A) Smooth	B) Distinctive	C) Large	D) Less intensity	
14	Few corners due to	A) High contrast	B) Excess smoothing	C) High resolution	D) Sharp edges	
15	Harris works well in	A) Uniform	B) Textured	C) Flat	D) Blurred	
16	Hough Transform is for	A) Noise removal	B) Shape detection	C) Compression	D) Color	
17	It works using	A) Ignoring noise	B) Voting	C) Reducing pixels	D) Grayscale	
18	Poor Hough result due to	A) Clean image	B) Noise	C) Strong edges	D) High resolution	
19	Hough maps to	A) Frequency	B) Parameter space	C) Colors	D) Edges	
20	Preprocessing helps Hough by	A) Adding noise	B) Enhancing edges	C) Reducing resolution	D) Removing features	
21	SIFT is invariant to	A) Noise	B) Scale & rotation	C) Color	D) Resolution	
22	Matching fails when	A) Similar images	B) Weak features	C) Low noise	D) High resolution	
23	SIFT works better due to	A) Color	B) Scale invariance	C) Size	D) Grayscale	
24	Feature matching is used in	A) Compression	B) Recognition	C) Noise removal	D) Histogram	
25	Incorrect matches due to	A) Strong features	B) Similar descriptors	C) High contrast	D) High resolution	
26	PCA is used for	A) Edge	B)	C) Noise	D) Smoothing	

		detection	Dimensionality reduction	addition		
27	PCA removes	A) Features	B) Correlation	C) Noise	D) Pixels	
28	PCA improves	A) Data size	B) Efficiency	C) Noise	D) Pixels	
29	PCA may cause	A) Data loss	B) Noise	C) Edge loss	D) Color change	
30	PCA is applied after	A) Segmentation	B) Feature extraction	C) Filtering	D) Thresholding	
31	Preprocessing + features improve	A) Noise	B) Accuracy	C) Size	D) Color	
32	Edge + Hough improves	A) Noise	B) Shape detection	C) Color	D) Compression	
33	Corner detection used for	A) Noise removal	B) Matching	C) Smoothing	D) Thresholding	
34	Feature extraction fails when	A) Noisy image	B) Clear image	C) Large image	D) Small image	
35	Preprocessing improves	A) Noise	B) Features	C) Pixels	D) Edges	
36	First step if edges fail	A) PCA	B) Preprocessing	C) SIFT	D) Hough	
37	Scale issue solved by	A) Sobel	B) SIFT	C) PCA	D) Thresholding	
38	Redundant features handled by	A) Edge	B) PCA	C) Hough	D) Smoothing	
39	Missing corners fixed by	A) Threshold	B) Size	C) Color	D) Resolution	
40	False Hough detection fixed by	A) Noise	B) Preprocessing	C) Color	D) Resolution	
41	Shape detection uses	A) PCA+SIFT	B) Edge+Hough	C) Smooth+PCA	D) Hist+SIFT	
42	Rotation matching uses	A) Sobel	B) SIFT	C) Threshold	D) PCA	
43	Feature extraction helps	A) Compression	B) Key points	C) Noise	D) Color	
44	Harris fails in	A) Textured	B) Uniform	C) Noisy	D) High contrast	
45	Preprocessing needed for	A) Visualization	B) Feature extraction	C) Compression	D) Storage	
46	Edge output used in	A) PCA	B) Hough	C) Smoothing	D) Threshold	
47	PCA transforms to	A) Spatial	B) Frequency	C) Feature space	D) Binary	

48	SIFT uses	A) Color	B) Gradients	C) Resolution	D) Size	
49	Matching depends on	A) Size	B) Distinctiveness	C) Color	D) Resolution	
50	Feature pipeline improves	A) Size	B) Accuracy	C) Noise	D) Compression	

Answer Key – MCQ Test (Computer Vision, ITA05)

Q.No	Correct Option	Answer
1	B	Reducing noise
2	C	Inconsistent
3	B	Reducing complexity
4	B	Foreground separation
5	B	Feature visibility
6	B	Intensity change
7	B	Low error
8	B	Noise
9	B	Low gradients
10	C	Noise dominates
11	C	Intensity variation
12	D	Noise
13	B	Distinctive
14	B	Excess smoothing
15	B	Textured
16	B	Shape detection
17	B	Voting
18	B	Noise
19	B	Parameter space
20	B	Enhancing edges
21	B	Scale & rotation
22	B	Weak features
23	B	Scale invariance
24	B	Recognition
25	B	Similar descriptors
26	B	Dimensionality reduction
27	B	Correlation
28	B	Efficiency
29	A	Data loss
30	B	Feature extraction
31	B	Accuracy
32	B	Shape detection
33	B	Matching
34	A	Noisy image
35	B	Features

Q.No	Correct Option	Answer
36	B	Preprocessing
37	B	SIFT
38	B	PCA
39	A	Threshold
40	B	Preprocessing
41	B	Edge+Hough
42	B	SIFT
43	B	Key points
44	B	Uniform
45	B	Feature extraction
46	B	Hough
47	C	Feature space
48	B	Gradients
49	B	Distinctiveness
50	B	Accuracy